CodeNest

CAPABILITIES / SOLUTIONS / DELIVERY

Software for the way your business works.

Service Catalog + Solution Handbook

Company knowledge base for clients and AI assistants

Web & mobile applications
AI, agents & knowledge assistants
ERP, automation & integrations
Cloud delivery & ongoing care

VERSION 1.0 | 21 SEPTEMBER 2026

Fictional company profile. All prices and timelines are illustrative estimates.

USD pricing throughout. Final terms require a project-specific proposal.

NAVIGATE THE HANDBOOK

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Reading guide: service details and pricing appear in Sections 03-13; FAQs in Section 20; practical project scenarios in Section 21. For RAG use, preserve section titles, table headers, USD units and estimate qualifiers with their content.

SECTION 01

Executive overview

CodeNest builds custom software for businesses that need clearer operations, better customer experiences and fewer manual tasks. Its services combine web and mobile applications, backend APIs, databases, AI, automation, integrations and cloud delivery.

Business challenge	CodeNest response
Work is spread across spreadsheets and inboxes	Centralize records, assign ownership and track progress in a business application or ERP.
Customers cannot self-serve	Build portals or mobile apps for requests, bookings, orders and status updates.
Teams repeat the same data entry	Connect existing systems and automate synchronization, approvals and notifications.
Company knowledge is hard to find	Build a permission-aware knowledge assistant with document search and cited answers.
An existing product is hard to maintain	Assess the code and infrastructure, then modernize it in stages.

A solution built around your workflow

CodeNest first maps the business process, users, data, current systems and expected outcome. It then recommends the smallest useful release and a stack that fits the work. A React web portal, a React Native app and a Python AI service can share the same backend and PostgreSQL data model when this makes the system easier to operate.

What clients can expect

A defined scope, visible milestones, regular demonstrations, testable acceptance criteria and documented handover. The proposed handover includes source code, deployment guidance, administrator notes and a prioritized improvement backlog; ownership and licensing terms are confirmed in the agreement.

> **Document status.** Version 1.0 | 21 September 2026. CodeNest is a fictional company in this demonstration knowledge base. Services and delivery practices are proposed company offerings. Prices, timelines and packages are illustrative estimates, not quotations or contractual commitments.

Use this handbook to shortlist a solution and prepare a consultation. No company address, email, phone number, customer history or certification has been supplied; this document does not invent those details.

SECTION 02

Service catalog

CodeNest offers standalone services and complete application delivery. The profiles on the following pages describe requirements, scope, delivery options and ongoing care. Detailed pricing tiers appear in the corresponding service chapters.

Service group	Primary deliverable	Detailed section
Applications	Web portals, mobile apps and custom business systems	03, 04, 17
Core platforms	Backend APIs, data models and ERP modules	05, 09, 10
Intelligent systems	AI features, agents and knowledge assistants	06, 07, 08
Connected operations	Automation and third-party synchronization	11, 12
Delivery and care	Cloud environments and ongoing maintenance	13, 22

Shared estimate basis

One-time project prices include requirements clarification, implementation, proportionate UI/UX work, testing, client review, documentation and the standard release setup described in the agreed scope. Standalone backend projects exclude a separate user interface; mobile estimates include a basic backend and light administration.

Third-party subscriptions, cloud usage, paid models, domains, store fees, taxes, major content creation, large data cleanup and ongoing support are excluded unless listed. A paid discovery phase is quoted separately only when substantial uncertainty prevents a reliable delivery scope.

> **Avoid double counting.** A full application package already includes its scoped backend, database and standard deployment. Do not add standalone service prices again. Additional services are priced only for the incremental work.

Estimated durations assume usable data and access at kickoff, one empowered client reviewer and feedback within two business days. Procurement, unavailable APIs, changing requirements, data cleanup and app-store review can extend elapsed time. Prices do not guarantee specific business outcomes.

Maintenance & Support - service profile

CodeNest can maintain new or existing applications through bug fixes, security and dependency updates, monitoring, database care and planned improvements. Clients provide repository and cloud access, incident history and priorities. The existing stack is assessed first; work is deployed through staging to the current host. Optional monitoring and help-desk integrations support incident tracking.

Size and estimate: Essential care includes up to 8 hours/month at $500-$900/month; Growth includes 20 hours at $1,200-$2,200/month; Managed includes 40 hours at $2,500-$4,500/month. Each month is a recurring service period; takeover effort is estimated after review. See Section 22 for scope and response targets.

SECTION 02 / CATALOG / 01-02

Service profiles

Web Application Development

Customer portals, SaaS products, dashboards, booking systems and admin tools. Solves scattered records, slow service and poor visibility.

Client inputs: A screen list, user roles, workflow examples, sample data and integration needs.

Recommended stack: React, TypeScript, Tailwind CSS; Node.js/Express; PostgreSQL, Drizzle and Zod.

Features and size: Login, role-based dashboards, forms, search, exports and notifications. Small: 6-10 screens. Larger: multiple teams, tenants or complex workflows.

Deployment: Vercel/Netlify frontend; Render, Railway or AWS backend; managed PostgreSQL. **Optional integrations:** Payments, CRM, email, calendars and storage.

Ongoing care: Dependency updates, access reviews, query performance and browser checks.

Estimated delivery: $6,000-$15,000; 4-8 weeks to $40,000-$100,000; 16-32 weeks across tiers.

Mobile Application Development

Android and iOS apps for customers and internal teams. Solves field access, on-the-go service and device-based workflows.

Client inputs: Target devices, app flows, offline needs, store accounts, branding and API access.

Recommended stack: React Native and TypeScript; Node.js APIs and PostgreSQL; managed push services.

Features and size: Login, profiles, orders, bookings, push notifications, camera uploads and optional offline queues. Simple: 6-10 screens. Complex: offline sync, live updates or specialized hardware.

Deployment: Google Play and Apple App Store; backend on Render, Railway or AWS. **Optional integrations:** Payments, maps, analytics, identity providers and business APIs.

Ongoing care: OS compatibility, store policy review, crash monitoring and push credentials.

Estimated delivery: $12,000-$25,000; 8-12 weeks to $55,000-$100,000; 20-32 weeks across tiers.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 02 / CATALOG / 03-04

Service profiles

Backend & API Development

Business APIs for web, mobile or partner systems. Solves inconsistent business rules and unreliable system-to-system access.

Client inputs: API consumers, entity definitions, roles, traffic assumptions and existing systems.

Recommended stack: Node.js, Express, TypeScript, PostgreSQL, Drizzle, Zod; Redis and Socket.IO when needed.

Features and size: REST endpoints, authorization, uploads, webhooks, jobs and audit events. Small: 8-15 endpoints. Complex: multi-tenant rules and queue-based processing.

Deployment: Render, Railway, DigitalOcean or AWS; managed PostgreSQL and object storage. **Optional integrations:** Identity, payments, CRM, messaging and partner APIs.

Ongoing care: API versioning, queue health, database tuning and secret rotation.

Estimated delivery: $4,000-$8,000; 3-5 weeks to $20,000-$60,000; 12-24 weeks across tiers.

AI & Machine Learning Solutions

AI features and applications for document intake, classification, recommendations and support. Solves slow information handling and repetitive review.

Client inputs: Representative examples, desired outputs, quality criteria, data permission and reviewers.

Recommended stack: Python/FastAPI or Node.js/TypeScript; OpenAI, Gemini or Anthropic; PostgreSQL.

Features and size: Extraction, summaries, classification, review queues, search and evaluation reports. One feature through a complete AI business application; custom model research scoped separately.

Deployment: Containerized API and workers on Render or AWS; managed database and file storage. **Optional integrations:** Existing apps, document stores, CRM and ticketing systems; OpenRouter if appropriate.

Ongoing care: Quality drift, model changes, prompts, usage budgets and review thresholds.

Estimated delivery: $3,000-$7,000; 2-4 weeks to $15,000-$40,000; 8-16 weeks across tiers.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 02 / CATALOG / 05-06

Service profiles

AI Agents

Bounded assistants that choose approved tools and execute multi-step work. Solves repetitive coordination across systems.

Client inputs: Allowed actions, approval rules, tool permissions, exception cases and success measures.

Recommended stack: LangGraph, optional LangChain, LLM APIs, Python or TypeScript, PostgreSQL.

Features and size: Lead triage, task creation, retrieval, draft replies, approvals and execution logs. One workflow with 2-3 tools through several workflows with 4-8 tools.

Deployment: Persistent workers on Render or AWS; database-backed checkpoints. **Optional integrations:** CRM, project tools, email, calendars, n8n and scoped database APIs.

Ongoing care: Tool changes, evaluation, action budgets, retries and permission review.

Estimated delivery: $10,000-$25,000; 6-10 weeks to $25,000-$60,000; 12-20 weeks across tiers.

RAG / Knowledge Base Applications

Searchable company knowledge with cited answers for staff or customers. Solves inconsistent answers and time spent locating documents.

Client inputs: Owned documents, access groups, update frequency, sample questions and source owners.

Recommended stack: React; Python/FastAPI or Node.js; PostgreSQL/pgvector or Supabase; embedding and LLM APIs.

Features and size: Uploads, ingestion status, vector search, citations, feedback and source administration. Starter: one workspace, up to 1,000 text documents; advanced: connectors and access groups.

Deployment: Vercel frontend, Render/AWS API and workers, managed PostgreSQL and S3. **Optional integrations:** Cloud drives, help desks, intranets and product documentation sources.

Ongoing care: Source freshness, deletion propagation, permissions, retrieval quality and token spend.

Estimated delivery: $8,000-$20,000; 6-10 weeks to $20,000-$45,000; 10-18 weeks across tiers.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 02 / CATALOG / 07-08

Service profiles

Database Development

Data models, schemas, migrations, optimization and vector search foundations. Solves duplicated records and slow or fragile data access.

Client inputs: Schema/export samples, data volume, business definitions and downtime limits.

Recommended stack: PostgreSQL, SQL, Drizzle migrations, pgvector; Python for data conversion.

Features and size: Constraints, indexes, migration scripts, reconciliation and backup configuration. 10-20 tables through multi-source data platforms and complex migrations.

Deployment: Neon, Supabase, AWS RDS or another managed PostgreSQL provider. **Optional integrations:** Legacy databases, CSV imports, applications, BI tools and external APIs.

Ongoing care: Backup checks, restore drills, capacity, access and query performance.

Estimated delivery: $2,000-$6,000; 2-4 weeks to $15,000-$30,000; 8-14 weeks across tiers.

ERP & Business Management Systems

Business modules linked around shared records. Solves disconnected inventory, purchasing, sales and operations.

Client inputs: Module priorities, approval paths, reports, sites, roles and migration samples.

Recommended stack: React, TypeScript, Node.js/Express, PostgreSQL, Drizzle; Redis for background jobs.

Features and size: Inventory, orders, purchasing, customers, suppliers, roles, reports and audit records. Basic: five core modules. Advanced: multiple departments/sites and integrations.

Deployment: Render or AWS; private-network AWS deployment when required. **Optional integrations:** Accounting, CRM, e-commerce, payroll and notification providers.

Ongoing care: Workflow changes, access reviews, release planning and data reconciliation.

Estimated delivery: $18,000-$35,000; 10-16 weeks to $70,000-$150,000; 24-40 weeks across tiers.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 02 / CATALOG / 09-10

Service profiles

Workflow Automation

Event-driven and scheduled workflows. Solves repeated data entry, missed handoffs and manual follow-ups.

Client inputs: Process maps, triggers, systems, credentials, volumes and exception owners.

Recommended stack: n8n, Node.js, APIs, webhooks, PostgreSQL and background jobs; AI where useful.

Features and size: Lead routing, onboarding, approvals, reminders, document intake and reports. 1-2 simple workflows through 6-10 workflows with complex exception handling.

Deployment: Managed n8n or hosted workers on Render, DigitalOcean or AWS. **Optional integrations:** CRM, email, calendars, forms, cloud storage and AI services.

Ongoing care: Credential renewal, failed job review, API changes and workflow ownership.

Estimated delivery: $2,000-$5,000; 2-4 weeks to $15,000-$30,000; 8-14 weeks across tiers.

Third-Party Integrations

API connectors and synchronization between existing products. Solves duplicate entry and inconsistent records across tools.

Client inputs: API documentation, sandbox accounts, field mappings, source-of-truth and volume.

Recommended stack: Node.js/TypeScript, REST APIs, webhooks, OAuth, PostgreSQL and job queues.

Features and size: Mapping, deduplication, incremental sync, retries, logs and reconciliation. One one-way entity flow through 3-5 connected systems with conflict rules.

Deployment: Client backend, managed workers on Render/Railway, or AWS. **Optional integrations:** CRM, accounting, payments, project tools, messaging, email, calendars and storage.

Ongoing care: Vendor API versions, expired credentials, rate limits and synchronization failures.

Estimated delivery: $1,500-$4,000; 1-3 weeks to $10,000-$25,000; 6-12 weeks across tiers.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 02 / CATALOG / 11-12

Service profiles

Custom Business Software

Bespoke case management, scheduling, field operations and approval systems. Solves workflows that standard products cannot adequately support.

Client inputs: Current process, business rules, screen priorities, roles and acceptance examples.

Recommended stack: Default web stack: React/TypeScript, Node.js/Express and PostgreSQL; mobile/AI by scope.

Features and size: Custom records, state transitions, task ownership, dashboards and exports. A focused operational application spanning several linked workflows.

Deployment: Vercel plus Render/Railway, or AWS for private networking and greater control. **Optional integrations:** Existing ERP, CRM, communications, document systems and partner services.

Ongoing care: Business rule changes, dependencies, monitoring and phased feature additions.

Estimated delivery: $15,000-$50,000; 8-20 weeks.

Cloud Deployment

Environment setup, release automation, monitoring and recovery planning. Solves risky manual releases and unclear operational ownership.

Client inputs: Repository, runtime needs, traffic assumptions, cloud account and recovery targets.

Recommended stack: Vercel/Netlify, Render/Railway/DigitalOcean/AWS, managed PostgreSQL, S3 and CI/CD.

Features and size: Environment separation, secret configuration, logs, alerts, backups and rollback. One app through a private-network environment with scaling and infrastructure as code.

Deployment: Client-owned cloud accounts; on-premises/hybrid feasibility assessed separately. **Optional integrations:** DNS, identity, monitoring, email delivery and existing networks.

Ongoing care: Capacity, provider costs, certificates, backups and deployment dependencies.

Estimated delivery: $1,000-$3,000; 1-2 weeks to $8,000-$20,000; 4-8 weeks across tiers.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 03

Web application development

CodeNest builds browser-based products for customers, staff and business partners. Typical projects include business dashboards, customer portals, admin panels, SaaS platforms, e-commerce applications, booking systems, CRM-style tools, operations platforms and data management applications.

Layer	CodeNest default	Project decision
Frontend	React, TypeScript, Tailwind CSS, reusable component library	Use an accessible component system; consider Next.js when server rendering or public search visibility matters.
Backend and data	Node.js, Express, TypeScript; PostgreSQL and Drizzle ORM	Centralize business rules and migrations. Add workers for long-running tasks.
Validation and identity	Zod; secure cookie sessions or carefully managed JWTs	Validate inputs on the server; enforce roles on every protected operation.
Deployment	Vercel or Netlify frontend; Railway, Render or AWS backend	Choose database region, networking and service capacity with the client.

Project tier	Typical scope	Est. timeline	Est. price (USD)
Small web application	6-10 screens, 2 roles, 3-5 core data entities	4-8 weeks	$6,000-$15,000
Medium web application	12-25 screens, 3-5 roles, 1-3 integrations	8-16 weeks	$15,000-$40,000
Large web application	Multiple workflows, tenants or complex reporting	16-32 weeks	$40,000-$100,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

How the scope grows

A small application might manage customer requests with two roles and a simple dashboard. A medium release adds approval stages, reports and integrations. A large release may include tenant isolation, advanced reporting, real-time collaboration or complex infrastructure.

Additional screens, roles, complex business rules, integrations, AI functions, real-time updates and advanced reporting increase design and test effort. Security requirements and infrastructure complexity can also change the architecture. A screen containing complex scheduling rules may require more work than several basic forms.

> **Typical delivery:** a responsive application, admin tools, documented API where required, versioned database migrations, test evidence and production handover. Payment processing, content migration and high-availability infrastructure are explicitly scoped.

SECTION 04

Mobile application development

CodeNest builds Android and iOS applications for customers, field teams and internal company use. React Native with TypeScript is the default proposal when the platforms share most business workflows and the required device capabilities are supported. The shared app connects to authenticated backend APIs and PostgreSQL.

Mobile project type	Typical scope	Est. timeline	Est. price (USD)
Simple mobile app	6-10 screens, login, profiles, one core workflow	8-12 weeks	$12,000-$25,000
Business mobile app	12-20 screens, bookings/orders, push, admin portal	12-20 weeks	$25,000-$55,000
Complex mobile app	Offline sync, real-time flows, deep device features	20-32 weeks	$55,000-$100,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Cross-platform or native?

Cross-platform development suits bookings, customer accounts, orders, staff checklists and business data capture. CodeNest would consider native Swift for iOS or Kotlin for Android when intensive graphics, specialized device SDKs, unusual background behavior or deep platform integration justify separate implementations. Native development is scoped separately; these prices assume a shared React Native application.

Component	Typical implementation
Mobile client	React Native, TypeScript, navigation, forms and secure device credential storage.
Backend	Node.js/Express, PostgreSQL, file storage and role-based APIs; basic backend included in these tiers.
User engagement	Push notifications through an appropriate provider, in-app status updates and deep links when scoped.
Operations	Crash/error reporting, a light admin interface and staged release configuration.

Store submission and release

CodeNest prepares signed Android and iOS builds, test releases, store metadata and submission support using client-owned Google Play and Apple developer accounts. The client supplies branding, policy text and business information and completes account verification. The release plan includes device testing and review feedback handling; store approval dates are outside CodeNest control.

Offline synchronization, payments, maps, localization, accessibility needs, device integrations and richer administration increase effort. A reusable, documented existing backend can reduce scope after assessment. Store fees and service usage are separate from development estimates.

Technology reference: React Native documentation supports Android and iOS development; the stack above is CodeNest's proposed delivery choice.

SECTION 05

Backend & API development

CodeNest builds the business layer behind web applications, mobile apps and partner integrations. The client receives an API contract, server implementation, database migrations, environment configuration, relevant automated tests and deployment guidance.

Capability	What CodeNest implements
REST APIs and validation	Versioned resources, documented requests/responses, Zod input checks and consistent errors.
Authentication and authorization	Login/session handling, roles, object ownership and tenant-level permission checks.
Business logic and data	Transactional rules, PostgreSQL persistence, Drizzle migrations and query indexes.
Files and processing	Upload limits, scoped object-storage access, background jobs and job status.
Real-time communication	Socket.IO for live notifications or collaboration where required; fallback and reconnect behavior.
External systems	Vendor API clients, signed webhook validation, retries and reconciliation.

Project tier	Typical scope	Est. timeline	Est. price (USD)
Small API backend	8-15 endpoints, 2 roles, one bounded workflow	3-5 weeks	$4,000-$8,000
Medium backend	20-40 endpoints, background jobs, 1-3 integrations	6-10 weeks	$8,000-$20,000
Complex backend platform	Multi-tenant workflows, queues, audit, scaling	12-24 weeks	$20,000-$60,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Default stack and deployment

Node.js, Express and TypeScript form the default backend. PostgreSQL stores business data, Drizzle manages typed queries and schema changes, and Zod validates external inputs. Redis is added for measured caching or queue needs. Socket.IO is optional, not a dependency for every application. Python/FastAPI may handle specialized AI processing alongside the main API.

Deploy to Render, Railway, DigitalOcean or AWS with managed PostgreSQL and object storage. Long-running processing is handled by workers rather than a request that must stay open. Persistent connections and worker requirements influence hosting selection.

> **Estimate boundaries:** endpoint counts indicate scope, not a per-endpoint price. Complex authorization or financial reconciliation can dominate the effort. These standalone prices exclude web/mobile user interfaces and extensive legacy data cleanup.

SECTION 06

AI & machine learning solutions

CodeNest adds AI to practical workflows: support chat, internal assistants, document intelligence, classification, content processing, search, recommendations and business automation. It can build a standalone product or integrate a focused feature into an existing application.

Project tier	Typical scope	Est. timeline	Est. price (USD)
Basic AI feature	One classification, extraction or summarization flow	2-4 weeks	$3,000-$7,000
AI chatbot	One chat channel, curated prompt flows, handoff	4-8 weeks	$6,000-$15,000
AI business application	Document/content workflow, review UI, analytics	8-16 weeks	$15,000-$40,000
Focused AI agent	One bounded workflow, 2-3 tools, approval step	6-10 weeks	$10,000-$25,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Need	Proposed approach	Client receives
Summaries, classification, content handling	LLM API through Node.js or Python/FastAPI	Structured output, validation, review controls and measured examples.
Document intelligence	Text extraction/OCR where needed, field extraction, business checks	Review queue, corrections, export and exception records.
Recommendations or prediction	Rules or a Python ML model depending on available data	Baseline comparison, evaluation and an API; training scope assessed first.
Chat and internal assistance	LLM API, conversation controls; RAG when company documents are required	Chat UI, escalation, usage limits and quality evaluation.

Model and architecture selection

CodeNest evaluates OpenAI, Google Gemini and Anthropic against representative client tasks. OpenRouter may be considered when multi-provider routing is useful and the client accepts its data path. LangChain can simplify model and retrieval integrations; LangGraph is useful for stateful workflows. PostgreSQL stores business records, with pgvector added for semantic retrieval.

The choice depends on measured answer quality, latency, usage cost, privacy, region availability and workflow complexity. A smaller model may handle classification while a stronger model handles difficult documents. CodeNest agrees review thresholds and test cases before release; AI output accuracy and business results are not guaranteed.

Chatbot estimates cover a focused chat experience. A document-backed knowledge assistant uses the RAG estimates in Section 08. Model training from scratch, large labeling programs and uncertain research work require separate discovery.

SECTION 07

AI agents

CodeNest builds agents that make bounded choices, select approved tools and carry out business tasks. A workflow can retrieve documents, inspect permitted records, call an API, create a task and request approval before a consequential action. The agent operates within explicit business permissions.

Example	Agent behavior	Control point
Lead processing	Read an inquiry, classify it, enrich permitted fields and propose a CRM task	Reviewer approves customer-facing outreach.
Operations assistant	Check order status, find relevant instructions and prepare the next task	Tool permissions restrict records and actions.
Document coordination	Retrieve supporting documents, compare required fields and route exceptions	Unclear or missing evidence enters a review queue.
Task management	Suggest priority and assign work using approved business rules	Assignment limits and a complete action history.

Project tier	Typical scope	Est. timeline	Est. price (USD)
Focused AI agent	One bounded workflow, 2-3 tools, approval step	6-10 weeks	$10,000-$25,000
Advanced AI agent system	Several workflows, 4-8 tools, recovery and audit	12-20 weeks	$25,000-$60,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Technical proposal

LangGraph can coordinate persistent workflow state, branching, approval pauses and recovery. Optional LangChain components connect models, retrieval and tools. CodeNest uses LLM function/tool calling with validated parameters; PostgreSQL stores state and audit events. Agents access business data through scoped service functions or APIs rather than unrestricted database credentials.

External APIs or n8n may execute deterministic steps. Persistent workers run on Render or AWS. The client receives configured workflows, a tool inventory, role/approval rules, execution logs, a review interface and a representative evaluation set.

> **Operating boundaries:** set maximum steps, time and spend per run; validate every tool request; prevent duplicate actions during retries; provide manual handoff. Payment, deletion, external messaging and sensitive record changes require the agreed approval policy.

More tools, complex recovery, long-running tasks and cross-department permissions increase effort. Model behavior, tool reliability and approval delays affect completion. Technology reference: LangGraph overview describes stateful orchestration and human oversight.

SECTION 08

RAG & knowledge base solutions

CodeNest builds knowledge assistants that answer questions using a client's approved documents and show source references. Typical products include internal company, HR, support, documentation and product assistants. Document search can also support legal teams, with professional review retained for interpretation.

Project tier	Typical scope	Est. timeline	Est. price (USD)
AI knowledge assistant	Up to 1,000 text documents, one workspace, citations	6-10 weeks	$8,000-$20,000
Advanced RAG system	Multiple sources, access groups, sync, re-ranking	10-18 weeks	$20,000-$45,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Product stage	What the client receives
Document intake	Upload interface, supported-format agreement, ownership metadata, import status and failure reports.
Processing and indexing	Text extraction, structured chunks, embedding generation and a searchable index.
Answering	Permission-filtered retrieval, optional re-ranking, LLM response and clickable document/page references.
Administration	Source updates, document deletion, feedback collection, access groups and usage reporting as scoped.

Recommended architecture

React presents chat and administration. A Python/FastAPI or Node.js service coordinates ingestion and answering. Original files live in object storage. PostgreSQL with pgvector stores metadata and vectors; Supabase is an alternative managed PostgreSQL platform. OpenAI or Gemini embeddings are selected after testing retrieval quality. LangChain is optional; LangGraph is added only for workflows that benefit from stateful orchestration.

The starter estimate assumes up to 1,000 mostly text PDF/DOCX/HTML documents, up to 10,000 pages total, one language, one workspace, common access permissions and manual uploads. Scanned pages/OCR, connector synchronization, per-document permissions, multilingual evaluation and large ingestion volumes move the work toward advanced scope.

> **Answer behavior:** cite supporting sources, distinguish missing evidence from a verified answer and offer human escalation. Test permission leakage, stale documents, conflicting versions and unsupported answers. Source documents are treated as data, not instructions that can grant tools or access.

The client owns source quality and approved access rules. CodeNest can maintain ingestion, retrieval tests and index refreshes under support. Reference: pgvector project documentation describes vector similarity search in PostgreSQL.

SECTION 09

Database development

CodeNest designs and improves the data foundations behind business software. Standalone database work is useful when an application already exists but data is duplicated, queries are slow, reporting is unreliable or a migration needs dedicated planning.

Project tier	Typical scope	Est. timeline	Est. price (USD)
Database foundation	10-20 tables, schema, migration, backup setup	2-4 weeks	$2,000-$6,000
Database optimization/migration	20-60 tables, profiling, cleansing, rehearsal	4-8 weeks	$6,000-$15,000
Complex data platform	Multiple sources, difficult migration, vector data	8-14 weeks	$15,000-$30,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Workstream	Deliverables and scope
Data and schema design	Business entity model, table relationships, constraints, naming conventions and a data dictionary.
Query optimization	Measured slow-query review, indexes, execution-plan analysis and before/after comparisons.
Migrations	Versioned scripts, field mapping, validation checks, rehearsal and a rollback/cutover plan.
Integration and vectors	Import/export processes, API access patterns and pgvector indexing where semantic search is needed.
Security and recovery	Least-privilege database roles, restricted network access, backup configuration and restore instructions.

SECTION 09

Data migration & operations

Default technology and alternatives

PostgreSQL is the default relational database. Drizzle migrations fit the TypeScript application stack; SQL and Python are used for specialized transformations and reconciliation. CodeNest can retain a suitable existing database after assessment instead of introducing migration work without a business reason. Vector indexing is scoped alongside permissions and metadata.

Data migration responsibilities

The client identifies record owners, business definitions, acceptable downtime and required retention. CodeNest profiles samples, agrees how duplicates and invalid records are handled, rehearses the import and checks counts and key totals. Large volumes, poor source quality and strict cutover windows increase the estimate.

Deployment options include Neon, Supabase, AWS RDS and managed PostgreSQL providers. Provider-specific extension support, backup retention and restore options are confirmed during planning. Backups require restore testing; neither migration work nor a managed database implies a zero-data-loss guarantee.

> **Separate from a full app:** standalone database prices cover the agreed data work. A new interface, complete business API, BI dashboard or major historical data cleanup is included only if named in the scope.

Migration review gates

Gate	Evidence before progressing
Sample review	Confirm source fields, ownership and duplicate/invalid-record rules.
Rehearsal	Load a safe copy, reconcile record counts and important business totals, and record runtime.
Cutover review	Confirm the write freeze or change-capture approach, fallback conditions and responsible owners.
Post-release check	Verify critical queries and imports, monitor exceptions and confirm the backup/restore procedure.

SECTION 10

ERP & business management

CodeNest builds ERP systems around the way a business buys, sells, holds inventory and coordinates work. Shared records connect departments, while roles and audit events show who changed information. Modules are selected for the client's priorities and delivered in phases.

ERP package	Typical scope	Est. timeline	Est. price (USD)
Basic ERP	Customers, products, inventory, orders, basic reports	10-16 weeks	$18,000-$35,000
Standard ERP	Purchasing, suppliers, approvals, more roles and reports	16-24 weeks	$35,000-$70,000
Advanced ERP	Multiple departments/sites, complex workflows, integrations	24-40 weeks	$70,000-$150,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Package	Included workflow examples	Stack and deployment
Basic ERP	Customers, products, stock movements, orders and basic reports; one site, 3 roles, clean CSV opening balances.	React/TypeScript, Node.js/Express, PostgreSQL/Drizzle; Render or AWS.
Standard ERP	Basic modules plus suppliers, purchasing, approval stages, staff records, 5 roles and up to 2 integrations.	Same stack, workers and Redis where needed; staging and production on Render or AWS.
Advanced ERP	Standard modules plus multiple departments/sites, complex approvals, operational automation, audit views and advanced reports.	Same core with queue processing; AWS private-network services and managed PostgreSQL when required.

Additional module choices

Employee records, notifications, supplier management, purchasing, finance-related approvals, payment status and operations tracking can be selected. CodeNest can connect an accounting or payroll platform. Statutory accounting, local tax calculation and payroll rules are separate requirements that need client domain review and explicit scope.

Rollout approach

Start with shared master data and one end-to-end workflow, such as purchase receipt through inventory to sales order. Rehearse migration, train module owners and pilot with one team before wider rollout. The client receives role definitions, operational reports, audit records, administrator training and migration reconciliation.

Custom valuation methods, manufacturing planning, barcode hardware, multiple legal entities, substantial historical data and specialized reporting increase cost and schedule. Modules are priced as one integrated scope; Basic, Standard and Advanced prices are alternatives, not amounts to add together.

SECTION 11

Workflow & process automation

CodeNest automates recurring operations so staff can focus on exceptions. Projects can route leads, onboard customers, send notifications, request approvals, assign tasks, synchronize records, process documents, manage email workflows and generate scheduled reports.

Project tier	Typical scope	Est. timeline	Est. price (USD)
Simple automation	1-2 workflows, 2 systems, notifications	2-4 weeks	$2,000-$5,000
Workflow automation system	3-5 workflows, 3 systems, retries, audit, approvals	4-8 weeks	$5,000-$15,000
Complex automation	6-10 workflows, custom jobs, complex exceptions	8-14 weeks	$15,000-$30,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Choice	Best fit	Delivery considerations
n8n	Connector-rich, moderate-volume workflows with business-readable steps	Agree hosting, credentials, workflow ownership, subscription/license fit and error handling.
Custom Node.js backend	Complex business rules, tenant isolation, strict performance or detailed testing needs	Use versioned code, database state, job queues, scheduled tasks and monitoring.
AI-assisted automation	Unstructured documents, incoming messages or classification before routing	Combine AI interpretation with deterministic validation and a human exception queue.

Example: lead-to-owner assignment

A form submission creates a normalized lead. The workflow checks duplicates, applies territory rules, writes to the CRM, notifies the assigned owner and schedules a follow-up reminder. A failed CRM call enters a retry queue; unresolved failures appear in an administrator view.

What clients receive

Documented workflow maps, triggers and schedules, configured connectors or custom code, credentials guidance, exception handling, execution logs and an operational handover. Acceptance checks include successful runs, invalid inputs, expired credentials and duplicate events.

> **Estimate boundaries:** workflow tiers include their listed connectors. Do not add the integration catalog price for those same connections. High volumes, complex approval branches, undocumented APIs and AI evaluation add scope.

Maintenance focuses on vendor API changes, failed executions, credential renewal, workflow revisions and cost/volume monitoring. The client appoints an owner who can resolve business exceptions.

SECTION 12

Third-party integrations

CodeNest connects existing products when replacing them would not solve the main problem. It can integrate CRM, accounting, project management, communication, payment, email, cloud storage, calendar and automation systems where supported APIs and permissions are available.

Project tier	Typical scope	Est. timeline	Est. price (USD)
Simple integration	One system pair, one-way sync, one entity	1-3 weeks	$1,500-$4,000
Medium integration	One system pair, two-way sync, 2-4 entities	3-6 weeks	$4,000-$10,000
Multi-system integration	3-5 systems, conflict rules, monitoring, reconciliation	6-12 weeks	$10,000-$25,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Integration concern	CodeNest delivery approach
Authentication	Use the vendor's supported OAuth or API-key flow; limit scopes and protect/rotate credentials.
Data mapping	Agree identifiers, field conversions, required values, duplicates and the authoritative system for each field.
Events and schedules	Use webhooks where available and scheduled incremental synchronization where needed.
Reliability	Validate webhook signatures, make writes idempotent, honor rate limits and retry transient failures.
Two-way synchronization	Define conflict rules, loop prevention, timestamps and deletion behavior before implementation.
Operations	Provide execution logs, failed-record review, replay/reconciliation and alert routing.

Typical delivery

The client receives a working connector, mapping specification, secret setup instructions, test evidence, runbook and a dashboard or logs appropriate to the project. CodeNest tests both initial transfer and ongoing changes, including API failures and partial updates.

Implementation usually uses Node.js/TypeScript, vendor APIs, webhooks, PostgreSQL for synchronization state and workers on the existing backend, Render, Railway or AWS. n8n can be suitable for straightforward connections with supported nodes.

> **Feasibility first:** access to a product does not necessarily include API access. The client must provide an eligible subscription, authorization and sandbox/sample records. Missing APIs, restrictive quotas or undocumented legacy protocols may require a revised approach.

Vendor subscriptions and transaction fees are excluded. Historical backfills, large file transfers, custom UI and poor data quality increase the estimate; simple one-way mappings and ready sandbox access can reduce it.

SECTION 13

Cloud & deployment

CodeNest selects hosting according to traffic, data location, security, reliability, scale, budget and client operations capacity. The proposed default is client-owned cloud accounts, separate secrets for each environment and a documented release and rollback process.

Layer	Deployment options	Selection guidance
Frontend	Vercel, Netlify, AWS hosting/CDN	Match static or server-rendered behavior, caching, region and deployment workflow.
Backend	AWS, Railway, Render, DigitalOcean	Confirm support for the runtime, persistent connections, workers and required network access.
Database	Neon, Supabase, AWS RDS, managed PostgreSQL	Choose region, extensions, connection limits, backup policy and recovery options.
File storage	AWS S3 or another suitable object-storage service	Use private buckets, controlled downloads, retention rules and upload restrictions.

Project tier	Typical scope	Est. timeline	Est. price (USD)
Basic deployment setup	One app, managed database, CI/CD, logs and backups	1-2 weeks	$1,000-$3,000
Production environment setup	Staging + production, workers, monitoring, recovery	2-4 weeks	$3,000-$8,000
Advanced infrastructure setup	Private networks, scaling, infrastructure as code	4-8 weeks	$8,000-$20,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Standard release deliverables

Environment configuration, automated build/deploy workflow, database migration procedure, health checks, log routing, basic alerting, backup configuration, a rollback plan and handover notes. More advanced environments can include infrastructure as code, private networks and controlled scaling.

Standard release setup is already included in an application package. These standalone estimates apply when CodeNest is engaged for deployment alone or for additional infrastructure beyond that package.

> **Ongoing costs are separate.** Hosting, storage, transfer, observability and AI usage are paid to providers or expressly included in a separate agreement. CodeNest proposes budgets and alerts; the client approves capacity and recurring spend.

An existing environment can be used after assessment. On-premises or hybrid deployment may be feasible, but hardware access, network constraints, maintenance ownership and availability of AI services need separate planning.

SECTION 13

Deployment configurations

These configurations are planning examples. User counts alone do not determine capacity: concurrent activity, background work, file sizes, database queries and availability targets matter more.

Configuration	Example topology	Illustrative monthly infrastructure budget
Small application	Vercel frontend; Render/Railway API; Neon/Supabase PostgreSQL; object storage; basic alerts.	USD $75-$250/month for a modest production workload.
Medium application	Frontend/CDN; API service and separate worker; managed PostgreSQL; optional Redis; staging; monitoring and backups.	USD $250-$1,000/month for moderate traffic and background processing.
Large application	AWS frontend/CDN; load-balanced container services; queue workers; managed PostgreSQL; S3; private networks; centralized monitoring.	USD $1,000-$4,000+/month depending on redundancy and workload.

How CodeNest sizes the environment

Record peak active users, request frequency, database size/growth, file storage/transfer, scheduled jobs, AI call rates and acceptable downtime. Run representative load tests where warranted, then configure a measurable baseline with thresholds for scaling. Confirm region and recovery requirements before selecting paid service tiers.

AI operating budget

An AI assistant adds model input/output usage, embeddings, optional re-ranking and document processing. CodeNest estimates monthly spend using expected requests multiplied by observed average usage per request, then adds ingestion and storage costs. Pilot measurements refine the budget before wider release; provider price changes can alter recurring cost.

Recovery and operational ownership

The proposal identifies who receives alerts, who can authorize changes and who pays cloud bills. Recovery time and acceptable data-loss windows are agreed per workload. Backup frequency, retention and restore testing follow those targets; high availability is a separately designed requirement.

> **Budget assumptions:** these are fictional planning allowances, not current provider quotations. They exclude development, maintenance retainers, taxes, AI APIs, messaging charges and store fees. Large media transfer, regional replication and intensive processing can exceed the ranges.

Reference: Render background worker documentation supports the proposed separation of background tasks from request-serving services.

SECTION 14

Security & authentication

CodeNest builds security controls into the agreed application scope and tests them against the system's roles, records and workflows. More demanding assurance, penetration testing or regulated deployment is scoped explicitly; this handbook does not claim certification or regulatory compliance.

Area	Proposed implementation	Client-visible result
Identity	Established authentication libraries; strong password hashing such as Argon2id where passwords are stored; reset and verification flows.	Reliable account lifecycle; SSO/MFA when scoped.
Sessions and JWTs	Secure, HttpOnly cookies where appropriate; CSRF protection for cookie-authenticated writes; expiry, rotation and revocation design.	Reduced credential exposure and controlled sign-out behavior.
Authorization	Server-side role, ownership and tenant checks; deny access by default.	Users see and change only permitted records.
API and input security	Zod server-side validation, rate limits, safe error responses and query parameterization.	Rejected invalid input and controlled resource usage.
Data and secrets	HTTPS; secret management; restricted database roles/network access; scoped file permissions.	Data and credentials are handled through documented controls.
Logging and reliability	Audit important events, redact secrets, monitor failures and define error/retry behavior.	Traceable operations without unnecessary sensitive data in logs.

AI and retrieval controls

Knowledge retrieval respects the requester's document access before content reaches the model. CodeNest designs source deletion and refresh behavior and tests for access leakage. Tools validate model-requested actions independently. Sensitive prompts and documents are sent only through approved providers and data-handling arrangements.

Shared responsibilities

CodeNest implements and documents the agreed controls. The client approves access roles, appoints administrators, defines retention and provides required policy/legal review. Provider configuration, hosting region, encryption and backup options are verified for the selected services at implementation.

> **Security deliverables:** access matrix, environment/secret guidance, relevant authorization tests, audit coverage and a prioritized risk list. Independent security assessment, formal compliance work and 24/7 incident response are separately agreed services.

SECTION 15

Development process

CodeNest moves from business problem to tested release through clear review points. The client appoints a decision maker and provides timely access, examples and feedback. Work is demonstrated in manageable increments, normally with a weekly status update and a review at each milestone.

Stage	CodeNest activity	What the client receives
01 Consultation	Discuss the business goal, users and current friction.	Initial fit assessment and next-step checklist.
02 Requirements	Map workflows, data, roles and external systems.	Requirements summary and open questions.
03 Scope definition	Prioritize the first release and identify exclusions.	Feature list, acceptance criteria and assumptions.
04 Technical planning	Check feasibility, dependencies and key risks.	Stack recommendation and implementation plan.
05 UI/UX planning	Design task flows, screens and states.	Wireframes or a prototype for review.
06 Architecture	Define components, data model and interfaces.	Architecture outline, role matrix and data model.
07 Development	Implement in small, reviewable releases.	Working demonstrations and progress updates.
08 Testing	Check business rules, integration and important failure paths.	Test results and a prioritized defect list.
09 Client review	Run agreed acceptance examples in staging.	Review log, fixes and release acceptance record.
10 Deployment	Release, verify health and complete handover.	Production access, runbook and release notes.
11 Maintenance	Monitor, fix and improve under an agreed plan.	Support reports and an improvement backlog.

Testing that matches the product

Coverage focuses on core business rules, permissions, critical user journeys and integration failures. Mobile projects add device checks; migrations add reconciliation; AI products add representative evaluation and unsupported-answer tests. Performance and security checks are sized to the agreed risk and workload.

> **Scope changes:** CodeNest explains the added work and any effect on price or schedule before implementing a requested change. Client review is an acceptance step; calendar dates remain estimates until dependencies are understood.

SECTION 16

Project estimation

CodeNest estimates work from a prioritized scope, representative data and technical constraints. Each feature is broken into design, implementation, integration, testing and release tasks. Dependencies and uncertainty determine the range and the proposed team capacity.

Estimate driver	What changes the effort
Users and roles	Concurrent activity, tenancy, permission rules, SSO and administrator responsibilities.
Features and business rules	Screens, workflow branches, validations, state transitions and unusual exceptions.
Integrations and data	Vendor API quality, field mapping, backfill size, duplicates and migration risk.
AI and automation	Evaluation data, tools, approval logic, document formats, source permissions and failure recovery.
Mobile and real time	Device coverage, offline sync, notifications, persistent connections and conflict handling.
Security and reporting	Audit depth, assurance requirements, reporting logic, retention and access isolation.
Deployment and support	Regions, recovery targets, traffic, staging, monitoring and support hours.

Illustrative additions to a scoped application

Incremental feature	Estimated additional price	Estimated additional time
One standard one-way CRM connection	$1,500-$4,000	1-3 weeks
One bounded AI summary feature	$3,000-$7,000	2-4 weeks
Basic live status updates for an existing workflow	$2,000-$5,000	1-3 weeks
Additional approval stage and associated reporting	$2,000-$6,000	1-3 weeks

Example: an agreed $12,000, 6-week small web application plus a $3,000 CRM connection and a $4,000 AI summary feature becomes an estimated $19,000 project. If these additions take 2 weeks each and run sequentially, the illustrative duration is 10 weeks. Parallel work can reduce elapsed time only when staffing and dependencies allow.

> **How a proposal is formed:** confirm scope and acceptance tests; estimate effort; identify dependencies; add an explicit uncertainty allowance; then agree milestones. A difficult legacy system may need paid discovery before a delivery estimate is credible.

All figures above are illustrative USD estimates. Final pricing depends on requirements; additions are charged only when not already included. Maintenance requirements influence design, but monthly support and provider bills remain separate.

SECTION 17

Sample project packages

These packages describe bounded starting points, not fixed-price commitments. Each one includes requirements clarification, proportionate design, development, testing, documentation and standard deployment. Choose one package for the same scope; included backend and database work is not charged again.

Example package	Typical scope	Est. timeline	Est. price (USD)
Business website	5-8 content pages, contact form, basic content editing	2-4 weeks	$2,000-$5,000
Small web application	6-10 screens, 2 roles, 3-5 core data entities	4-8 weeks	$6,000-$15,000
AI knowledge assistant	Up to 1,000 text documents, one workspace, citations	6-10 weeks	$8,000-$20,000
Basic ERP	Customers, products, inventory, orders, basic reports	10-16 weeks	$18,000-$35,000
Workflow automation system	3-5 workflows, 3 systems, retries, audit, approvals	4-8 weeks	$5,000-$15,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Package	Client receives	Key boundary
Business website	Responsive frontend, contact form, basic CMS/content structure, standard deployment.	Client supplies copy/branding; excludes customer accounts and custom business workflows.
Business web application	Authentication, dashboard, backend, PostgreSQL, 2 roles and CRUD for 3-5 entities.	6-10 screens; no large migration or complex integration.
AI knowledge assistant	Document upload, vector search, RAG chat and source references.	One workspace; up to 1,000 text documents and 10,000 pages; manual source updates.
Custom ERP - Basic	Customers, products, inventory, orders, user roles and basic reports.	One site; clean opening-data import; advanced modules quoted separately.
Workflow automation system	3-5 workflows, APIs/webhooks, notifications, up to 3 systems and execution logs.	Includes scoped connectors; vendor subscriptions excluded.

Custom business software

A bespoke operational system that does not fit these packages is estimated at $15,000-$50,000; 8-20 weeks. Examples include case management, scheduling and field operations with linked workflows. CodeNest selects a web, mobile or combined interface after reviewing how the team actually works.

Payment schedules, ownership terms, warranty arrangements and support commitments are specified in the project proposal. No such commercial terms are fixed by these illustrative packages.

SECTION 18

Technology stack reference

This matrix is CodeNest's proposed starting point, not a requirement to use every listed technology. Existing client systems, team preferences and verified constraints can justify alternatives. Supported stable versions and provider compatibility are confirmed during technical planning.

Area	Preferred technologies	Alternatives / typical use
Frontend	React, TypeScript, Tailwind CSS, reusable components	Next.js when server rendering or public content needs justify it; portals and dashboards.
Backend	Node.js, Express, TypeScript	Python/FastAPI for AI/data-heavy services; business APIs and jobs.
Mobile	React Native, TypeScript	Swift/iOS or Kotlin/Android for deep platform needs; customer and field apps.
Database	PostgreSQL	Retain an appropriate existing database after review; business records and reporting.
ORM / migrations	Drizzle ORM and versioned migrations	Direct SQL for complex queries and optimization.
Validation	Zod in TypeScript services	Python validation for FastAPI services; API and form input checks.
AI	OpenAI, Google Gemini or Anthropic APIs	OpenRouter for agreed multi-provider routing; Python ML for data-backed prediction.
RAG	Retrieval service; optional LangChain	LangGraph when multi-step state is needed; grounded document assistants.
Vector search	PostgreSQL + pgvector	Supabase PostgreSQL or an assessed dedicated vector service for special scaling needs.
Automation	n8n; Node.js workers and webhooks	Custom job processing for complex rules, control and isolation.
Cloud	Vercel/Netlify; Render/Railway; managed PostgreSQL	AWS/DigitalOcean for suitable workloads; region and operations requirements drive choice.
Authentication	Secure cookie sessions; established auth libraries	JWTs for appropriate API clients; managed identity/SSO when scoped.
Real time / queues	Socket.IO; Redis when justified	Polling for modest update needs; queue technology chosen for workload.
File storage	AWS S3 or suitable object storage	Existing approved cloud storage; private uploads and controlled downloads.
Monitoring	Structured logs, health checks and error tracking	Sentry, provider monitoring/CloudWatch, OpenTelemetry where suitable; alert and incident visibility.

Named products are technology options only. Their inclusion does not imply a partnership, certification or endorsement. Licensing, service availability, data handling and commercial terms are checked for the project.

SECTION 19

Service comparison

Start with the outcome, then choose the service. Several services can share one solution: a mobile client can use a custom backend, an ERP can include automation, and a knowledge assistant can appear inside an existing portal.

Choose	Business problem it solves	Best fit / scope boundary
Website	People need to understand the business and make contact.	Content pages, forms and basic editing; no substantial operational workflow.
Web application	Users need accounts and browser-based tasks.	Portals, dashboards, bookings and business data management.
Mobile application	Users need frequent on-the-go or device-based access.	Android/iOS tasks, push and device features; app-store delivery.
ERP	Departments need shared operational records and controls.	Inventory, purchasing, orders, roles, reporting and audit.
AI application	Staff must process unstructured text or documents.	Extraction, summaries, classification or recommendations with review.
AI agent	A task requires choosing tools and coordinating actions.	Bounded multi-step work with approval and recovery rules.
RAG system	Answers must be grounded in approved company knowledge.	Document retrieval, citations, source updates and permissions.
Workflow automation	Repetitive steps and handoffs follow known rules.	Triggers, schedules, approvals, notifications and exception handling.
Custom backend	An existing or new interface needs reliable business logic.	APIs, authorization, data handling and background jobs.
Third-party integration	Existing tools need to exchange consistent records.	Field mapping, sync, retries and reconciliation between systems.

Common selection decisions

Chatbot or knowledge assistant? A focused conversational flow fits the chatbot tier. Answers from company documents require retrieval, source management and citation scope. **Automation or agent?** Use explicit rules for predictable work; add an agent only where flexible interpretation and tool selection are useful.

Custom system or existing software? CodeNest reviews whether configuration or integration of the current tools can meet the need. Custom software is appropriate when business rules, user experience or data control cannot be met adequately by the available system.

> **For a first release:** select one high-value workflow, the minimum roles and the necessary integrations. Validate actual usage before expanding into additional modules, mobile apps or AI actions.

SECTION 20 / FAQ / 01-08

Customer questions & answers

Answers below use the same scope and illustrative USD estimates as the service chapters. Final pricing and timelines depend on a project-specific proposal.

01. What services does CodeNest provide?

CodeNest builds web and mobile applications, backend APIs, AI/ML solutions, agents, RAG assistants, databases, ERP systems, automation, integrations and custom business software. It also provides cloud deployment and maintenance. See Sections 02-13 and 22.

02. How much does a web application cost?

CodeNest estimates a small web application at $6,000-$15,000, medium at $15,000-$40,000 and large at $40,000-$100,000. These illustrative USD ranges depend on screens, roles, business rules and integrations. See Section 03.

03. How long does a mobile application take?

CodeNest estimates 8-12 weeks for a simple mobile app, 12-20 weeks for a business app and 20-32 weeks for a complex app. Store review and client dependencies may extend elapsed delivery. See Section 04.

04. Can CodeNest build both Android and iOS apps?

Yes. CodeNest proposes a shared React Native application for Android and iOS when requirements fit cross-platform delivery, with platform-specific testing and store submissions. Native development can be assessed for specialized needs. See Section 04.

05. What technology does CodeNest use for mobile apps?

The default proposal is React Native with TypeScript, connected to Node.js/Express APIs and PostgreSQL. Push notifications, file storage, authentication and monitoring are added according to scope. See Section 04.

06. Can CodeNest build an ERP system?

Yes. CodeNest estimates Basic ERP at $18,000-$35,000 over 10-16 weeks, Standard at $35,000-$70,000 over 16-24 weeks and Advanced at $70,000-$150,000 over 24-40 weeks. Modules are selected around the client workflow. See Section 10.

07. Can CodeNest integrate our existing CRM?

Yes, subject to supported APIs, suitable account permissions and a workable data mapping. A simple one-way integration is estimated at $1,500-$4,000 over 1-3 weeks; two-way or multi-system work has larger scope. See Section 12.

08. Can CodeNest connect multiple software systems?

Yes. CodeNest can synchronize 3-5 systems with field mapping, conflict rules, retries and reconciliation. The multi-system tier is estimated at $10,000-$25,000 over 6-12 weeks. See Section 12.

SECTION 20 / FAQ / 09-16

Customer questions & answers

Answers below use the same scope and illustrative USD estimates as the service chapters. Final pricing and timelines depend on a project-specific proposal.

09. Can CodeNest build an AI chatbot?

Yes. A focused AI chatbot is estimated at $6,000-$15,000 over 4-8 weeks. If the chatbot must answer from company documents, the RAG scope in Section 08 is the relevant estimate. See Section 06.

10. Can CodeNest build a company knowledge assistant?

Yes. The starter AI knowledge assistant includes upload, retrieval, chat and citations at an estimated $8,000-$20,000 over 6-10 weeks. Advanced access groups and source connectors use the advanced RAG tier. See Section 08.

11. What database does CodeNest normally use?

PostgreSQL is the default for business records. Drizzle supports TypeScript queries and migrations, and pgvector can support semantic retrieval. An appropriate existing database can be retained after review. See Sections 09 and 18.

12. Where can CodeNest deploy our application?

Frontend options include Vercel, Netlify and AWS. Backends can use AWS, Railway, Render or DigitalOcean. Databases can use Neon, Supabase, AWS RDS or other managed PostgreSQL providers. See Section 13.

13. Can CodeNest deploy on AWS?

Yes. AWS is an option for frontends, APIs, workers, managed PostgreSQL and S3 storage, including private-network environments where required. Capacity, recovery and operational ownership are planned for the project. See Section 13.

14. Can CodeNest use Vercel?

Yes. Vercel is a proposed frontend platform. Dedicated APIs, persistent connections or long-running workers can be hosted separately when required by the architecture. See Sections 03 and 13.

15. Can CodeNest build custom APIs?

Yes. CodeNest provides REST APIs, authentication, authorization, business rules, files, jobs and integrations. Standalone backend estimates range from $4,000-$8,000 for small scope to $20,000-$60,000 for complex platforms. See Section 05.

16. Can CodeNest automate our business processes?

Yes. CodeNest can automate leads, onboarding, approvals, notifications, data sync, email and reporting using n8n or custom workers. Estimated tiers range from $2,000-$5,000 to $15,000-$30,000. See Section 11.

SECTION 20 / FAQ / 17-24

Customer questions & answers

Answers below use the same scope and illustrative USD estimates as the service chapters. Final pricing and timelines depend on a project-specific proposal.

17. Can CodeNest maintain our application after launch?

Yes. Example plans are Essential at $500-$900/month, Growth at $1,200-$2,200/month and Managed at $2,500-$4,500/month. Included capacity and response targets are defined in Section 22; provider bills are separate.

18. What affects the final project price?

Features, users, roles, business rules, integration difficulty, data quality, AI evaluation, mobile/offline needs, real-time behavior, reporting, security and infrastructure all affect effort. CodeNest confirms assumptions in a proposal. See Section 16.

19. Can an existing application be modernized?

Yes. CodeNest can review the repository, dependencies, architecture, database and deployment, then propose phased upgrades or replacement of specific components. The estimate depends on the findings and target scope. See Sections 01, 09 and 22.

20. Can CodeNest integrate AI into an existing application?

Yes. A bounded summary, extraction or classification feature is estimated at $3,000-$7,000 over 2-4 weeks when the existing system is ready for integration. Larger AI workflows use the AI application tier. See Section 06.

21. Can CodeNest build around our existing workflow?

Yes. CodeNest maps roles, process stages, exceptions and data before recommending the solution. Custom business software is estimated at $15,000-$50,000 over 8-20 weeks for a focused operational application. See Sections 02 and 17.

22. What is included in a project estimate?

The agreed scope normally includes requirements clarification, proportionate design, development, testing, review, documentation and standard release setup. Hosting, paid APIs, subscriptions and ongoing support are separate unless expressly included. See Section 02.

23. Do we pay separately for the backend in a full application?

Only for additional scope. A full web/mobile package includes its scoped backend and database, so the standalone backend price should not be added again. Mobile tiers include a basic backend and light administration. See Sections 02 and 04.

24. Are prices and delivery dates fixed?

No. All handbook prices and timelines are illustrative estimates for the fictional company. Requirements, access, reviews and technical dependencies determine the final proposal; CodeNest does not promise fixed delivery dates in this document. See Sections 01 and 16.

SECTION 20 / FAQ / 25-32

Customer questions & answers

Answers below use the same scope and illustrative USD estimates as the service chapters. Final pricing and timelines depend on a project-specific proposal.

25. Which AI model will CodeNest choose?

CodeNest evaluates OpenAI, Gemini and Anthropic using client examples and considers cost, latency, privacy and complexity. OpenRouter may be considered for agreed multi-provider routing. Exact models are selected during planning. See Section 06.

26. Can an AI agent send messages or change business records?

It can perform approved actions through scoped tools. The project defines approvals for external messaging, payments, deletion and sensitive updates, along with validation, action limits and logs. See Section 07.

27. Will the knowledge assistant always give a correct answer?

No. CodeNest tests retrieval and answers, requires supporting citations and designs missing-evidence responses and escalation. Document quality and model behavior affect reliability, so accuracy is not guaranteed. See Section 08.

28. Can a knowledge assistant restrict access by department?

Yes, through scoped access groups and permission-filtered retrieval. Multiple access groups and per-document permission needs belong in the advanced RAG scope, estimated at $20,000-$45,000 over 10-18 weeks. See Section 08.

29. Can CodeNest process scanned documents?

Yes, OCR and validation can be scoped for document intelligence or RAG. The starter RAG estimate assumes mostly text documents; scan quality, languages and layout variation affect added effort. See Sections 06 and 08.

30. Can CodeNest migrate our existing data?

Yes. CodeNest can profile sources, map fields, rehearse migrations, validate results and plan cutover. Cleanup needs, volume and downtime constraints affect the estimate. Standalone data tiers appear in Section 09.

31. Can our mobile app work offline?

Offline storage and synchronization can be scoped. Conflict rules, queued changes, device testing and connectivity failures increase effort and usually move the app toward the complex tier. See Section 04.

32. Will CodeNest handle Google Play and App Store submission?

CodeNest prepares builds, test releases, metadata and submission support. The client supplies developer accounts and completes verification. Store approval timing and fees are outside development estimates. See Section 04.

SECTION 20 / FAQ / 33-40

Customer questions & answers

Answers below use the same scope and illustrative USD estimates as the service chapters. Final pricing and timelines depend on a project-specific proposal.

33. Can CodeNest build finance workflows in our ERP?

Yes. Purchasing approvals, payment status and finance-related processes can be included. Statutory accounting, tax and payroll rules require separate requirements and client domain review. See Section 10.

34. What is the difference between n8n and custom automation?

CodeNest proposes n8n for connector-rich, moderate-volume flows and custom Node.js workers for complex business logic, isolation or performance requirements. The choice depends on scope and maintenance ownership. See Section 11.

35. Can CodeNest use our current cloud account?

Yes. Client-owned accounts are the proposed default, and an existing environment can be assessed. CodeNest confirms permissions, network access, capacity, secret handling and responsibility for provider bills. See Section 13.

36. What should we prepare for the first consultation?

Bring the problem, users, current process, must-have features, sample data, existing tools, integration needs, budget range and desired launch window. Identify a decision maker and any security or hosting constraints. See Section 23.

37. What happens if we request new features during development?

CodeNest describes the added work and effect on cost or schedule before implementing the change. Features already included are not charged again. An updated scope or later release can be agreed. See Sections 15 and 16.

38. Do we receive source code and documentation?

The proposed handover includes source code, deployment guidance, administrator notes and an improvement backlog. Ownership, licenses and any third-party components are confirmed in the agreement. See Sections 01 and 15.

39. Does CodeNest offer 24/7 support?

The example support plans do not include 24/7 cover or guaranteed resolution times. Coverage hours, timezone and incident commitments are confirmed in the support agreement; round-the-clock cover requires a separate arrangement. See Section 22.

40. How can we contact CodeNest for a proposal?

Use the contact channel through which CodeNest supplied this handbook or request a verified consultation channel from its representative. This fictional knowledge base contains no verified email address, phone number or website. See Section 24.

SECTION 21 / ILLUSTRATIVE SCENARIOS / 01-02

Client project scenarios

01. Retail inventory and orders

Client problem: A single-site retailer tracks products and stock in separate spreadsheets, leading to uncertain availability and repeated order entry.

CodeNest solution: CodeNest builds a Basic ERP with one shared product and stock ledger.

Recommended stack: React/TypeScript, Node.js/Express, PostgreSQL and Drizzle.

Major features: Customers, products, receipts/issues, order status, low-stock alerts and basic reports; 3 roles.

Deployment: Frontend plus API on Render/AWS and managed PostgreSQL; private file storage if required.

Estimated scope and delivery: Basic ERP; $18,000-$35,000; 10-16 weeks.

Estimate factors: One site and clean opening stock fit the baseline. Barcode hardware, multiple branches and historical stock correction increase effort; omitting alerts can narrow it.

02. Internal operations dashboard

Client problem: Managers collect status updates from separate teams and cannot see who owns delayed work.

CodeNest solution: CodeNest builds a browser-based operations platform with shared task states and reporting.

Recommended stack: React, Tailwind CSS, TypeScript, Node.js/Express, PostgreSQL.

Major features: 12-18 screens, 3 roles, assigned tasks, due dates, a manager dashboard, exports and one notification integration.

Deployment: Vercel frontend, Render API, managed PostgreSQL and a small reminder worker.

Estimated scope and delivery: Medium web application; $15,000-$40,000; 8-16 weeks.

Estimate factors: Ready process definitions keep work within the medium tier. Complex report logic and extra departments increase effort; fewer roles and integrations can reduce it.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 21 / ILLUSTRATIVE SCENARIOS / 03-04

Client project scenarios

03. Customer booking mobile app

Client problem: Customers phone to make bookings and staff manually send updates or reschedule visits.

CodeNest solution: CodeNest builds shared Android/iOS booking apps and a light administration interface.

Recommended stack: React Native/TypeScript, Node.js APIs, PostgreSQL and push notifications.

Major features: 12-20 screens, accounts, service catalog, availability, bookings, status updates and admin scheduling.

Deployment: Google Play and Apple App Store; APIs on Render or AWS; managed PostgreSQL.

Estimated scope and delivery: Business mobile app; $25,000-$55,000; 12-20 weeks.

Estimate factors: Live dispatch, payments and offline editing add scope. Existing tested scheduling APIs can reduce backend work. Store review can extend elapsed delivery.

04. Automated lead processing

Client problem: Leads arrive from forms and email; staff copy them to CRM and often miss follow-up deadlines.

CodeNest solution: CodeNest automates lead intake, assignment and reminder workflows across three systems.

Recommended stack: n8n or Node.js workers, CRM/email APIs, webhooks and PostgreSQL run state.

Major features: 3-5 workflows, deduplication, owner assignment, notifications, retry handling and failed-record review.

Deployment: Managed n8n or a worker on Render/DigitalOcean; protected connector credentials.

Estimated scope and delivery: Workflow automation system; $5,000-$15,000; 4-8 weeks.

Estimate factors: Clear routing rules and supported connectors fit the tier. AI lead qualification, more channels and unusual CRM fields increase effort; a form-only flow can reduce it.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 21 / ILLUSTRATIVE SCENARIOS / 05-06

Client project scenarios

05. Internal knowledge assistant

Client problem: Staff search folders and repeatedly ask colleagues for policy and product information.

CodeNest solution: CodeNest builds a single-workspace assistant over approved internal documents with source references.

Recommended stack: React, FastAPI, PostgreSQL/pgvector, OpenAI or Gemini embeddings and an approved LLM API.

Major features: Upload, import status, semantic retrieval, cited answers, feedback and source management.

Deployment: Vercel frontend, Render API/worker, managed PostgreSQL and S3 originals.

Estimated scope and delivery: AI knowledge assistant; $8,000-$20,000; 6-10 weeks.

Estimate factors: Assumes common access and up to 1,000 text documents/10,000 pages. OCR, department permissions and drive sync increase scope; fewer sources simplify it.

06. Growing distributor ERP

Client problem: Sales, purchasing and warehouse teams each keep separate records and approvals are hard to audit.

CodeNest solution: CodeNest builds a Standard ERP around purchasing through fulfillment, with phased department rollout.

Recommended stack: React/TypeScript, Node.js/Express, PostgreSQL/Drizzle and background workers.

Major features: Basic ERP modules plus suppliers, purchasing, approval stages, 5 roles, staff records and up to 2 integrations.

Deployment: Staging and production on AWS or Render, managed PostgreSQL and object storage.

Estimated scope and delivery: Standard ERP; $35,000-$70,000; 16-24 weeks.

Estimate factors: Reconciled master data and one clear process keep scope bounded. Multiple entities, manufacturing and extensive history increase effort; fewer modules lower scope.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 21 / ILLUSTRATIVE SCENARIOS / 07-08

Client project scenarios

07. CRM and accounting connection

Client problem: Teams retype customer details in CRM and accounting, creating inconsistent addresses and duplicate records.

CodeNest solution: CodeNest builds two-way synchronization for 2-4 entities with field ownership and conflict rules.

Recommended stack: Node.js/TypeScript, vendor OAuth APIs, PostgreSQL sync state and a worker queue.

Major features: Field mapping, initial bounded transfer, updates, deduplication, failure alerts and reconciliation.

Deployment: Worker on the client backend or Render/AWS, with controlled credentials and logs.

Estimated scope and delivery: Medium integration; $4,000-$10,000; 3-6 weeks.

Estimate factors: Usable APIs and clean identifiers fit the tier. Large historical backfills and custom tax mappings add scope; one-way customer sync reduces it.

08. Document-processing AI

Client problem: An operations team manually reads varied supplier documents and enters key fields into a system.

CodeNest solution: CodeNest builds a document intake application with extraction and a human review queue.

Recommended stack: React, Python/FastAPI, OCR where required, approved LLM API, PostgreSQL and S3.

Major features: Upload, field extraction, business validation, confidence/review routing, correction history and export.

Deployment: API and document worker on Render/AWS; managed database and private file storage.

Estimated scope and delivery: AI business application; $15,000-$40,000; 8-16 weeks.

Estimate factors: Assumes a limited agreed document family and representative samples. Many layouts, handwriting or multilingual extraction increase scope; text-only files reduce it.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 21 / ILLUSTRATIVE SCENARIOS / 09-10

Client project scenarios

09. Custom SaaS platform

Client problem: A business wants to sell its service to multiple customer companies with separate accounts and administration.

CodeNest solution: CodeNest builds a scoped multi-tenant SaaS release with organization-level access isolation.

Recommended stack: React/TypeScript, Node.js/Express, PostgreSQL/Drizzle, job queues and payment APIs.

Major features: Tenant onboarding, roles, subscription integration, core business workflow, reporting and audit events.

Deployment: Frontend/CDN, API and workers on AWS; managed PostgreSQL, private S3 and monitoring.

Estimated scope and delivery: Large web application; $40,000-$100,000; 16-32 weeks.

Estimate factors: Complex billing, per-tenant customization, SSO and advanced availability increase scope. One standardized workflow and fewer roles reduce effort.

10. Synchronize three business systems

Client problem: Orders, CRM records and fulfillment status differ across an e-commerce platform, CRM and warehouse tool.

CodeNest solution: CodeNest creates a synchronization service with a source of truth for each field and reconciliation views.

Recommended stack: Node.js/TypeScript, webhooks, scheduled jobs, PostgreSQL and queue-backed workers.

Major features: 3-system mapping, incremental sync, duplicate protection, retry/replay, conflict handling and alerts.

Deployment: Dedicated worker and API on AWS or Render, managed database and operational monitoring.

Estimated scope and delivery: Multi-system integration; $10,000-$25,000; 6-12 weeks.

Estimate factors: High order volume, missing webhooks and complex cancellations add effort. Well-documented APIs and limited historical transfer can reduce it.

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

SECTION 22

Maintenance & support

CodeNest can maintain applications it builds and assess existing products for takeover. Ongoing work covers bug fixes, security and dependency updates, performance improvements, small feature additions, database maintenance, monitoring and infrastructure care. AI products can add model compatibility reviews, quality evaluation and RAG source/index updates.

Example plan	Included monthly capacity	Illustrative price (USD)	Coverage focus
Essential	Up to 8 hours	$500-$900/month	Prioritized fixes, dependency checks and basic health review.
Growth	Up to 20 hours	$1,200-$2,200/month	Essential care plus performance/data work and small improvements.
Managed	Up to 40 hours	$2,500-$4,500/month	Broader application/infrastructure care, release coordination and AI/RAG reviews if scoped.

Example response target	Essential	Growth	Managed
Critical: production unusable	Within 1 business day	Within 4 business hours	Within 2 business hours
Normal: non-blocking issue	Within 2 business days	Within 1 business day	Within 1 business day

Response targets mean initial assessment, not guaranteed resolution. They apply only during agreed coverage hours. Working days, hours, timezone, channels and escalation contacts are defined in the support agreement. The example plans do not include 24/7 coverage.

How maintenance capacity is used

Investigation, fixes, testing, release work, meetings and maintenance tasks use the included hours. Large features, major migrations and substantial new integrations receive a separate estimate. Additional capacity requires agreement; rollover, overage rates and minimum terms are specified in the proposal rather than assumed here.

Taking over an existing application

CodeNest reviews the repository, infrastructure, dependencies, known incidents and access before committing to support. Required stabilization or setup is separately estimated. The monthly plan begins when responsibilities and the operating baseline are agreed.

> **Price depends on complexity and support needs.** Hosting, AI/API usage, vendor licenses and independent security assessments are excluded. A defect warranty, if offered, is defined separately in the project agreement; this handbook does not promise a warranty period.

SECTION 23

Client onboarding

After a potential client contacts CodeNest, the first objective is to understand the business problem and whether a practical first release can address it. CodeNest then turns that discussion into a scope and a project-specific proposal.

Step	What happens	Output
Initial discussion	Identify the problem, users, business impact and desired outcome.	Fit assessment and a preparation checklist.
Business analysis	Map the current process and where it fails.	Workflow summary and success measures.
Requirements	Collect roles, features, sample data and integration needs.	Prioritized requirements and dependencies.
Feasibility	Review APIs, legacy systems, AI samples and hosting constraints.	Technical options, risks and discovery needs.
Scope and estimate	Define the first release, exclusions, milestones and assumptions.	Estimated price and delivery range.
Proposal	Document deliverables, acceptance and commercial/support terms.	Reviewable project-specific proposal.
Planning and kickoff	Confirm responsibilities, access, reviewers and the first milestone.	Delivery backlog and kickoff plan.

Prepare for the first consultation

• A short statement of the problem and the result you want to measure, such as reducing duplicate entry or enabling online booking.

• Your current workflow, user groups, roles, existing tools and must-have versus later features.

• Representative documents or sample records, API documentation and an overview of data quality. Use anonymized samples when appropriate.

• Preferred budget range, desired launch window, constraints, required integrations and existing cloud or app-store accounts.

• Security, access, hosting region, retention, availability and maintenance expectations, plus one empowered decision maker.

> **You do not need to choose a stack first.** CodeNest recommends the technology after reviewing the business workflow, existing systems and delivery constraints. Unknowns are recorded and resolved through targeted discovery, not hidden in a promised date.

SECTION 24

Final service summary

CodeNest combines application development, AI, data, integrations and operations into a scoped solution. The table below lists representative entry packages; larger tiers and scope boundaries are documented in the service chapters.

Business need	CodeNest solution	Typical technologies	Est. timeline	Est. price (USD)
Public presence	Business website	React, content editing, Vercel	2-4 weeks	$2,000-$5,000
Customer/staff portal	Small web application	React, Node.js, PostgreSQL	4-8 weeks	$6,000-$15,000
Android/iOS access	Simple mobile app	React Native, APIs, PostgreSQL	8-12 weeks	$12,000-$25,000
Application logic	Small API backend	Node.js, Express, Drizzle	3-5 weeks	$4,000-$8,000
AI content processing	Basic AI feature	LLM API, Python/Node.js	2-4 weeks	$3,000-$7,000
Task coordination	Focused AI agent	LangGraph, tools, PostgreSQL	6-10 weeks	$10,000-$25,000
Company knowledge	Starter RAG assistant	pgvector, embeddings, LLM API	6-10 weeks	$8,000-$20,000

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Combine services into one scoped solution

These entry packages address different needs. A complete application can combine interface, API, database, automation and deployment work in one proposal. Refer to each service chapter for larger tiers, inputs and exclusions.

SECTION 24 / FINAL SERVICE SUMMARY / CONTINUED

Delivery, care & consultation

Business need	CodeNest solution	Typical technologies	Est. timeline	Est. price (USD)
Reliable data	Database foundation	PostgreSQL, SQL, migrations	2-4 weeks	$2,000-$6,000
Shared operations	Basic ERP	React, Node.js, PostgreSQL	10-16 weeks	$18,000-$35,000
Repeatable processes	Simple automation	n8n, APIs, webhooks	2-4 weeks	$2,000-$5,000
Connected tools	Simple integration	Node.js, APIs, OAuth	1-3 weeks	$1,500-$4,000
Bespoke workflows	Custom business software	React, Node.js, PostgreSQL	8-20 weeks	$15,000-$50,000
Production release	Basic deployment setup	Cloud host, CI/CD, database	1-2 weeks	$1,000-$3,000
Ongoing care	Essential support	Existing stack, monitoring	Monthly service	$500-$900/month

All amounts are illustrative USD estimates. Timelines are estimated elapsed weeks after kickoff; final scope, price and schedule require a project-specific proposal.

Discuss your project with CodeNest

Share your business problem, current systems, priority features and budget range through the contact channel that supplied this handbook. CodeNest can then review feasibility, recommend a first release and prepare a proposal with scope, estimated cost, timeline, deployment and support options.

> **Consultation contact:** request the verified contact channel from the CodeNest representative or the application hosting this knowledge base. No verified email, phone number, address or website is supplied in this fictional company profile; an assistant should not invent one.